Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)			It's the time taken to halve (1) the activity / count rate / mass / number of {atoms / nuclei} (1)	2			2		
	(b)			Throw the cubes (1) Count the number of cubes that have landed with the shaded face upwards **OR** count the rest (1) Remove the shaded cubes **and** repeat these steps [many times] (1)	3			3		3
	(c)	(i)		Reduces the effect of randomness Accept it is more accurate Don't accept reference to reproducible / repeatable / reliable	1			1		1
		(ii)	I	Correct construction lines vertically and horizontally that meet the curve (1) Half-life = 1.7 ± 0.1 [throws] (1) Accept answers of 2 [throws] if correct construction lines are present		2		2		2
			II	Ticks in boxes 1, 3 and 6 i.e. 1F cubes show decay at the slowest rate 3F cubes show the equivalent of two half-lives after 2 throws The number of 2F cubes remaining after 4 throws is about $\frac{1}{5}$ of the original Deduct 1 mark for each additional tick			3	3		3

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			75 hours = 5 half-lives (1) 48 $\rightarrow$ 24 $\rightarrow$ 12 $\rightarrow$ etc (1) sequencing of halving starting from 48 Answer = 1.5 [mg] (1) **OR** $\frac{1}{2^5}$ (1) $\frac{1}{32}$ (1) Answer = 1.5 [mg] (1)		3		3	3	
				Question 6 total	**6**	**5**	**3**	**14**	**3**	**9**